

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 06.1

Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB.

Name: Dhinakaran. S **Reg. No.:** 192412353

Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
%% Signal
fs = 1000; f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);
%% Quantization Levels
L = [8 16 32];
figure;
for i = 1:length(L)
    % Uniform quantization (manual formula expressed with valid MATLAB syntax)
    xq{i} = 2*round((x+1)/2)*(L(i)-1)/(L(i)-1) - 1;
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);
    subplot(4,1,i+1);
    stairs(t,xq{i},'LineWidth',1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    grid on;
end
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
% Plot Quantized Signals
subplot(4,1,i+1);
stairs(t,xq{i},colors{i},'LineWidth',1.2);
title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
grid on;
end

% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;

% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));

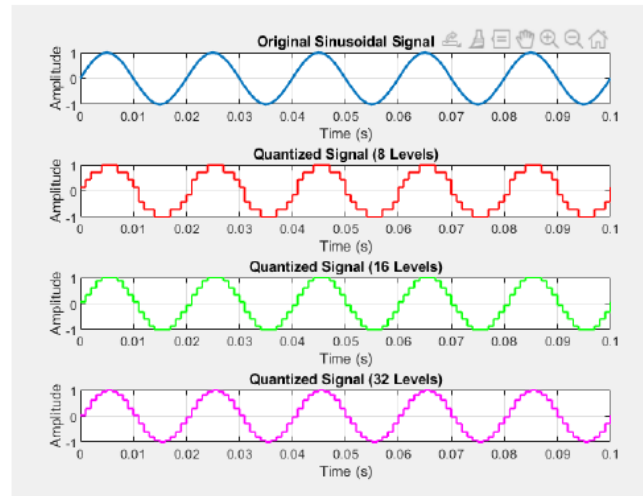
%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end
```

Output:

Objective 1:

Input:

- Sampling Frequency, $F_s = 1000\text{Hz}$
- Signal Frequency, $f = 50\text{Hz}$
- Time Duration = 0.1 s



Result:

1. A discrete-time sinusoidal signal was successfully generated in MATLAB.
2. Uniform quantization was performed for different quantization levels, and the corresponding quantized signals were obtained.
3. Comparison of the original and quantized signals showed that the quantized signal approximates the original signal more closely as the number of quantization levels increases.
4. The quantization error was analyzed and found to decrease with increasing quantization levels, resulting in improved signal quality and reduced quantization noise.

Practical Question 1:

Generate a discrete-time sinusoidal signal in MATLAB with amplitude $A = 1$, frequency $f = 100\text{Hz}$, and sampling frequency $F_s = 1000\text{Hz}$. Perform uniform quantization using 8

quantization levels. Plot the original signal, quantized signal, and quantization error. Analyze the effect of quantization on the signal quality.

Task:

1. Generate the sinusoidal signal.
2. Implement uniform quantization with 8 levels.
3. Plot the original and quantized signals.
4. Determine and plot the quantization error.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2:

Generate a discrete-time sinusoidal signal in MATLAB with amplitude $A = 2$, frequency $f = 200\text{Hz}$, and sampling frequency $F_s = 2000\text{Hz}$. Perform uniform quantization using 16 and 32 quantization levels. Compare the quantized signals and analyze the variation in quantization error for the two cases.

Task:

1. Generate the sinusoidal signal.
2. Quantize the signal using 16 and 32 levels.
3. Compare the original and quantized signals.
4. Compute and analyze the quantization errors for both quantization levels.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To study uniform quantization at 8, 16 and 32 levels and analyze quantization error and MSE.

Student Details

Name: Dhinakaran. S

Reg. No.: 192412353

Course: ECA09 - Digital Signal Processing